Class 6 Science – Chapter 6: Materials Around Us

इस अध्याय के दिए गए 8 पृष्ठों के आधार पर नीचे सभी दिखाई देने वाले Quick Check, गतिविधियों और प्रश्नों के उत्तर सरल अंग्रेज़ी में दिए गए हैं। इसके बाद हर मुख्य topic से महत्वपूर्ण और संभावित परीक्षा-प्रश्नों के उत्तर भी शामिल हैं।

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1. Places in My Neighbourhood (Page 77)

The pictures show a sweater, trousers, a wolf, a lion, flowers, a bird and a peacock. We can group them according to their common characteristics.

Name of the group	Examples
Clothes	Sweater, trousers
Animals	Wolf, lion
Flowers	Sunflower, hibiscus
Birds	Small bird, peacock

Q1. Why do we group objects into different categories?

Answer: We group objects to arrange them systematically and find them easily.

Q2. On what basis can objects be classified?

Answer: Objects can be classified by their properties, origin, material, size, shape, colour, use and availability.

2. Classification of Materials (Page 78)

Q1. What is a material?

Answer: Any substance used to make an object is called a material.

Q2. What is classification?

Answer: Classification is the process of sorting and grouping things on the basis of certain criteria.

Q3. Why is classification important?

Answer: Classification helps us arrange objects properly and find a particular item easily.

Q4. Give examples of classification based on origin.

Answer: Fibres can be classified as natural fibres and synthetic fibres.

Q5. How can materials be classified on the basis of their properties?

Answer: Materials can be classified on the basis of appearance, texture, hardness, solubility, density, transparency and conductivity of heat and electricity.

Q6. Give an example of classification based on the material used to make an object.

Answer: Crockery can be classified into earthenware and stoneware.

3. Appearance: Lustrous and Non-lustrous Materials (Page 78)

Q1. What is lustre?

Answer: Lustre is the shiny appearance of a material.

Q2. What are lustrous materials? Give examples.

Answer: Materials that appear shiny are called lustrous materials. Gold, silver and many metals are examples.

Q3. What are non-lustrous materials?

Answer: Materials that appear dull are called non-lustrous materials. Wood, chalk and some plastics are examples.

Q4. Why is gold used to make jewellery?

Answer: Gold is lustrous and has a shiny appearance, so it is used to make jewellery.

Q5. Give two examples each of lustrous and non-lustrous materials.

Answer:

- Lustrous: Gold and silver.
- Non-lustrous: Wood and chalk.

Activity: Shiny and Dull Objects

Q. Write five examples each of shiny and dull objects found at home.

Answer:

- Shiny objects: Steel spoon, stainless-steel plate, silver jewellery, metal key, aluminium utensil.
- Dull objects: Wooden table, chalk, cardboard, brick, unpolished wood.

4. Texture (Page 79)

Q1. What is texture?

Answer: Texture is how the surface of a material feels when we touch it.

Q2. What are rough materials?

Answer: Rough materials have uneven surfaces with bumps and ridges. Examples are tree bark, stone and brick.

Q3. What are smooth materials?

Answer: Smooth materials have even surfaces without noticeable bumps and ridges. Examples are paper, glass and plastic sheets.

Q4. How can we identify the texture of a material?

Answer: We can gently touch its surface and feel whether it is rough or smooth.

5. States of Matter (Page 79)

Q1. What is matter?

Answer: Anything that has mass and occupies space is called matter.

Q2. Name the three states of matter.

Answer: The three states of matter are solid, liquid and gas.

Q3. What are solids?

Answer: Solids have a fixed shape and a fixed volume. Their particles are closely packed, so they are not easily compressed.

Examples: Ice, stone and wood.

Q4. What are liquids?

Answer: Liquids have a fixed volume but no fixed shape. They take the shape of the container in which they are kept.

Examples: Water, milk and cooking oil.

Q5. What are gases?

Answer: Gases have neither a fixed shape nor a fixed volume. They are highly compressible because their particles are loosely packed.

Examples: Air, oxygen and water vapour.

Q6. Why does water take the shape of its container?

Answer: Water is a liquid and does not have a fixed shape.

Q7. Why are gases highly compressible?

Answer: Gas particles are loosely packed and have large spaces between them.

Q8. What is the difference between a solid, a liquid and a gas?

Property	Solid	Liquid	Gas
Shape	Fixed	Not fixed	Not fixed
Volume	Fixed	Fixed	Not fixed
Compressibility	Very low	Very low	High
Example	Ice	Water	Water vapour

6. Hardness (Pages 79–80)

Q1. What is hardness?

Answer: Hardness describes how difficult it is to cut, scratch or compress a material.

Q2. What are hard materials?

Answer: Hard materials are difficult to compress and cannot be cut easily.

Examples: Stone, metal keys and iron nails.

Q3. What are soft materials?

Answer: Soft materials can be compressed or cut more easily.

Examples: Sponge, rubber and cotton.

Q4. How can a harder material affect a softer material?

Answer: A harder material can cut or compress a less hard material. For example, a knife can cut vegetables and fruits.

Q5. Which is the hardest naturally occurring substance on Earth?

Answer: Diamond.

7. Solubility (Page 80)

Q1. What is solubility?

Answer: Solubility is the property of a substance to dissolve in a solvent.

Q2. What is a soluble substance?

Answer: A substance that dissolves in a liquid is called soluble in that liquid.

Examples: Salt and sugar dissolve in water.

Q3. What is an insoluble substance?

Answer: A substance that does not dissolve in a liquid is called insoluble in that liquid.

Examples: Sand and chalk powder are insoluble in water.

Q4. What is a solvent?

Answer: A solvent is a substance, usually a liquid, in which another substance dissolves.

Q5. What is a solution?

Answer: A solution is a uniform mixture formed when a substance dissolves in a solvent.

Q6. What are miscible liquids?

Answer: Miscible liquids mix well with each other to form a uniform mixture.

Examples: Alcohol and water. The textbook also lists milk and water as an example.

Q7. What are immiscible liquids?

Answer: Immiscible liquids do not mix uniformly with each other.

Example: Cooking oil and water.

Q8. Why is water called a universal solvent?

Answer: Water is called a universal solvent because it dissolves many substances. However, it does not dissolve every substance.

Q9. Why can aquatic animals survive in water?

Answer: Oxygen dissolved in water is used by aquatic animals for breathing.

Q10. Do all gases dissolve in water?

Answer: No. Some gases, such as oxygen and carbon dioxide, dissolve in water, while gases such as hydrogen and nitrogen are described in the textbook as not dissolving in water.

In the Lab: Solubility Activity (Page 81)

The activity asks students to test substances such as salt, sugar, chalk powder, sand and cooking oil in water.

Substance	Observation in water
Salt	Soluble
Sugar	Soluble
Chalk powder	Insoluble
Sand	Insoluble
Cooking oil	Immiscible with water

Q. What is the conclusion of this activity?

Answer: Different substances behave differently in water. Some dissolve, while others do not dissolve or mix uniformly.

8. Density (Page 81)

Q1. What is density?

Answer: Density is the mass of a substance per unit volume.

Q2. How does density affect floating and sinking?

Answer: An object less dense than water generally floats, while an object denser than water generally sinks.

Q3. Name three materials that may float on water.

Answer: A plastic ball, cooking oil and an air-filled balloon.

Q4. Name three materials that sink in water.

Answer: An iron nail, a stone and a brick.

Q5. Why does an iron nail sink in water?

Answer: Iron is denser than water, so an iron nail sinks.

Q6. Why does cooking oil float on water?

Answer: Cooking oil is less dense than water.

Lab Activity: Layers of Liquids (Page 81)

The picture shows different liquids arranged in layers according to their densities.

Q1. Which substance has the lowest density in the pictured arrangement?

Answer: Rubbing alcohol, shown at the top of the liquid layers.

Q2. Which substance has the highest density among the liquids shown?

Answer: Honey, shown at the bottom of the liquid layers.

Q3. Which has a lower density: maple syrup or dish soap?

Answer: Dish soap.

Q4. Which is the most dense liquid in the pictured arrangement?

Answer: Honey.

Q5. Is honey miscible with water?

Answer: Yes, honey can mix with water, although it mixes slowly because it is thick.

Note: These answers follow the labelled arrangement in the textbook illustration.

9. Volume (Page 81)

Q1. What is volume?

Answer: Volume is the amount of space occupied by an object.

Q2. What is the capacity of a container?

Answer: Capacity is the amount of liquid or other substance a container can hold.

Q3. What is the difference between volume and capacity?

Answer: Volume is the space occupied by an object. Capacity is the amount a container can hold.

Q4. Why does a football have more volume than a cricket ball?

Answer: A football occupies more space than a cricket ball.

Q5. What are the common units of volume or capacity mentioned in this chapter?

Answer: Litre (L) and millilitre (mL) are units used for liquid volume or capacity.

10. Quick Check: True or False (Page 82)

1. Grouping things into different categories helps us locate things easily.
Answer: True

2. We can classify chalk and plastic on the basis of their appearance.
Answer: True

3. Lustrous metals can be used to make jewellery.
Answer: True

4. Sugar is insoluble in water.
Answer: False. Sugar is soluble in water.

11. Transparency (Page 82)

Q1. What is transparency?

Answer: Transparency is the property of a material that determines how much light can pass through it.

Q2. What are transparent materials?

Answer: Materials that allow light to pass through completely are called transparent materials.

Examples: Clear glass, air and water.

Q3. What are translucent materials?

Answer: Materials that allow only some light to pass through them are called translucent materials.

Examples: Butter paper, wet muslin cloth and frosted glass.

Q4. What are opaque materials?

Answer: Materials that do not allow light to pass through them are called opaque materials.

Examples: Wood, a metal sheet, a brick wall and a notebook.

Q5. What is the difference between transparent, translucent and opaque materials?

Type	How light passes through	Examples
Transparent	Almost all light passes through	Clear glass, air
Translucent	Some light passes through	Butter paper, frosted glass
Opaque	Light does not pass through	Wood, brick

Activity: Testing Transparency

Q. What do you observe when you look at a cross mark through clear, lightly coloured and black-painted glass slides?

Answer: The cross mark can be seen clearly through clear glass, less clearly through lightly coloured glass, and cannot be seen through black-painted opaque glass.

Conclusion: Materials differ in the amount of light they allow to pass through them.

12. Conduction of Heat and Electricity (Page 83)

Q1. What is conduction?

Answer: Conduction is the transfer or flow of heat or electricity through a material.

Q2. What are good conductors of heat?

Answer: Materials that allow heat to pass through them easily are called good conductors of heat.

Examples: Iron and aluminium.

Q3. What are insulators of heat?

Answer: Materials that do not allow heat to pass through them easily are called insulators of heat.

Examples: Plastic, wood and glass.

Q4. What are good conductors of electricity?

Answer: Materials that allow electric current to flow through them easily are called good conductors of electricity.

Examples: Most metals, such as copper and aluminium.

Q5. What are electrical insulators?

Answer: Materials that do not allow electric current to flow through them easily are called electrical insulators.

Examples: Plastic, air and wood.

Q6. Why are cooking utensil handles made of plastic or wood?

Answer: Plastic and wood are poor conductors of heat. They reduce the transfer of heat to our hands and help prevent burns.

Q7. Are all materials good conductors of heat and electricity?

Answer: No. Metals are generally good conductors, while materials such as plastic and wood are poor conductors.

13. Magnetic Materials (Page 83)

Q1. What are magnetic materials?

Answer: Materials that are attracted by a magnet are called magnetic materials.

Q2. Give examples of magnetic materials.

Answer: Iron, steel and many objects made from them, such as iron nails and steel pins.

Q3. Are all metals magnetic?

Answer: No. Not all metals are attracted to a magnet. Iron and steel are common examples of magnetic materials.

Q4. How can we test whether an object is magnetic?

Answer: Bring a magnet close to the object. If the object is attracted to the magnet, it is magnetic.

Q5. Name two common non-magnetic materials.

Answer: Wood and plastic.

14. Matter, Mass and Units (Pages 83–84)

Q1. What is matter?

Answer: Anything that has mass and occupies space is called matter.

Q2. Give five examples of matter.

Answer: Air, water, trees, humans and food are examples of matter.

Q3. What are the two common properties of all materials mentioned in the chapter?

Answer: Mass and volume.

Q4. What is mass?

Answer: Mass is the amount of matter in an object.

Q5. What is the unit of mass?

Answer: Mass is measured in kilograms (kg) and grams (g).

Q6. What is the unit of volume or liquid capacity?

Answer: Volume or liquid capacity is commonly measured in litres (L) and millilitres (mL).

Q7. Are living things and non-living things made of matter?

Answer: Yes. Both living and non-living things are made of matter.

15. Properties of Commonly Used Materials (Page 84)

The textbook summarises the properties of wood, metals, plastic and glass. The following table is based on that summary.

Property	Wood	Metals	Plastic	Glass
Appearance	Dull	Shiny	Dull or shiny	Shiny

Hardness	Hard	Hard	Hard or soft	Hard
Texture	Rough	Smooth	Smooth	Smooth
Solubility in water	Insoluble	Insoluble	Insoluble	Insoluble
Transparency	Opaque	Opaque	Opaque or translucent	Transparent or translucent
Conduction of heat and electricity	Poor conductor	Mostly good conductor	Poor conductor	Poor conductor
Density compared with water, as stated in the text	Lower	Higher	Lower	Higher

Q1. Which materials are generally good conductors of heat and electricity?

Answer: Metals are generally good conductors of heat and electricity.

Q2. Which materials are poor conductors of heat and electricity?

Answer: Wood, plastic and glass are poor conductors of heat and electricity.

Q3. Why is glass used to make windows?

Answer: Clear glass is transparent, so it allows light to pass through and helps us see outside.

Q4. Why is plastic used to make many household objects?

Answer: Plastic can be hard or soft, is generally insoluble in water and is a poor conductor of heat and electricity.

Q5. Why are metals used to make cooking utensils?

Answer: Metals are generally good conductors of heat, so they transfer heat to the food.

Q6. Why is wood used to make some furniture?

Answer: Wood is strong and hard enough for making many types of furniture.

16. Quick Check: Fill in the Blanks (Page 84)

1. The materials through which light passes easily are called transparent materials.
2. Aluminium is a good conductor of heat.
3. Butter paper is an example of a translucent material.
4. Metals are good conductors of heat and electricity.

17. Additional Important Questions and Answers

ये अतिरिक्त प्रश्न chapter के सभी प्रमुख concepts की परीक्षा-तैयारी में मदद करेंगे।

A. Classification and Properties

Q1. Why can the same material be used to make different objects?

Answer: A material may have properties that make it suitable for different uses.

Q2. Can we classify an object in more than one way?

Answer: Yes. An object can be classified by its material, colour, shape, hardness, transparency or use.

Q3. Why is classification useful in a supermarket?

Answer: Products are arranged in groups, so customers can find them easily.

Q4. What is the difference between lustrous and non-lustrous materials?

Answer: Lustrous materials look shiny, while non-lustrous materials look dull.

Q5. What is the difference between rough and smooth materials?

Answer: Rough materials have uneven surfaces, while smooth materials have even surfaces.

B. Solubility and Density

Q1. What happens when salt is mixed with water and stirred?

Answer: Salt dissolves in water and forms a solution.

Q2. What happens when sand is mixed with water?

Answer: Sand does not dissolve in water and may settle at the bottom.

Q3. What happens when cooking oil and water are mixed?

Answer: They do not mix uniformly. The oil forms a separate layer, usually above the water.

Q4. Why can two objects of similar size behave differently in water?

Answer: They may have different densities, which affect whether they float or sink.

Q5. Does every object that floats have a lower mass than every object that sinks?

Answer: No. Floating and sinking depend on density and the object's interaction with the liquid, not simply on total mass.

C. Transparency and Conductivity

Q1. Why can we see through a clean glass window but not through a wooden door?

Answer: Clear glass is transparent, while wood is opaque.

Q2. Why can we see light through butter paper but not see objects clearly through it?

Answer: Butter paper is translucent. It allows some light to pass through but scatters the light.

Q3. Why should electric wires have insulating coverings?

Answer: Insulating coverings help prevent accidental contact with the conducting wire and reduce the risk of electric shock.

Q4. Why should we not touch electrical appliances with wet hands?

Answer: Water containing dissolved substances can conduct electricity, increasing the risk of electric shock.

D. Matter, Mass and Volume

Q1. How are mass and volume different?

Answer: Mass tells us how much matter an object contains, while volume tells us how much space it occupies.

Q2. Does air occupy space?

Answer: Yes. Air occupies space and has mass, so it is matter.

Q3. Give two examples each of solids, liquids and gases.

Answer:

- Solids: Stone and ice.
- Liquids: Water and milk.
- Gases: Oxygen and air.

Q4. Why does a liquid take the shape of its container?

Answer: A liquid has no fixed shape, so it takes the shape of its container.

Q5. Why does a gas fill the available space in a container?

Answer: Gas particles are loosely packed and move freely, so a gas spreads to fill the available space.

18. Multiple-Choice Questions (MCQs)

Test Yourself

0/12 answered

1. What is classification?

A. Mixing all objects together
B. Grouping things by certain criteria
C. Changing the shape of an object
D. Measuring temperature

2. Which of these is a lustrous material?

A. Chalk
B. Wood
C. Silver
D. Brick

3. Which is an example of a translucent material?

A. Clear glass
B. Butter paper
C. Wooden board
D. Metal sheet

4. Which substance dissolves in water?

A. Sand
B. Chalk powder
C. Sugar
D. Cooking oil

5. Which pair of liquids is immiscible?

A. Water and alcohol
B. Water and cooking oil
C. Water and water
D. Milk and water, as described in the text

6. Which is a good conductor of heat?

A. Plastic
B. Wood
C. Aluminium
D. Air

7. Which material is attracted to a magnet?

A. Wood
B. Plastic
C. Iron
D. Glass

8. What is the amount of space occupied by an object called?

A. Mass
B. Volume
C. Texture
D. Lustre

9. Which state of matter has a fixed shape and volume?

A. Solid
B. Liquid
C. Gas
D. Vapour

10. Which material is generally opaque?

A. Air
B. Clear glass
C. Wood
D. Clear water

11. Which unit is used to measure mass?

A. Litre
B. Kilogram
C. Metre
D. Second

12. Why are wooden or plastic handles used on many cooking utensils?

A. They are good conductors of heat
B. They are poor conductors of heat
C. They dissolve in water
D. They are transparent
Check Answers

19. Fill in the Blanks: Practice

1. Anything that has mass and occupies space is called matter.
2. The property of a material that describes its shiny appearance is called lustre.
3. A substance that dissolves in water is called soluble in water.
4. Liquids that do not mix uniformly are called immiscible liquids.
5. The amount of space occupied by an object is called volume.
6. Materials that allow light to pass through completely are called transparent materials.
7. Materials that allow only some light to pass through are called translucent materials.
8. Materials that do not allow light to pass through are called opaque materials.
9. Materials that allow heat to pass through easily are called good conductors of heat.
10. Materials attracted by a magnet are called magnetic materials.

20. True or False: Practice

1. All materials have the same properties. — False
2. Salt dissolves in water. — True
3. Cooking oil and water mix uniformly. — False
4. Gases have a fixed shape and volume. — False
5. Clear glass is transparent. — True
6. Wood is generally a poor conductor of heat. — True
7. All metals are attracted to magnets. — False
8. Mass and volume are properties of matter. — True
9. A translucent material allows some light to pass through. — True
10. A substance that does not dissolve in water is insoluble in water. — True

21. One-Line Revision

- Material: A substance used to make an object.
- Classification: Grouping objects by certain criteria.
- Lustrous: Shiny in appearance.
- Texture: How a surface feels.
- Soluble: Able to dissolve in a particular solvent.
- Miscible liquids: Liquids that mix uniformly.
- Density: Mass per unit volume.
- Volume: Space occupied by an object.

- Capacity: Amount a container can hold.
- Transparent: Allows light to pass through completely.
- Translucent: Allows some light to pass through.
- Opaque: Does not allow light to pass through.
- Conductor: Allows heat or electricity to pass through easily.
- Insulator: Does not allow heat or electricity to pass through easily.
- Magnetic material: A material attracted by a magnet.
- Matter: Anything that has mass and occupies space.

Exam tip: Learn the definitions, examples, differences between properties, and the textbook Quick Check answers. The most important topics in this chapter are classification, states of matter, solubility, density, transparency, conductivity, magnetic materials, mass and volume.